

Airmar® NMEA 2000® / NMEA 0183 Bootloader Specification

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- DRAFT -

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1. Introduction

This document describes the operation of the flash bootloader module within the Airmar® NMEA 2000® / NMEA 0183 family of products. The primary purpose of the bootloader is to provide a means to perform field upgrades to the firmware stored in the flash memory.

Features of the flash bootloader are as follows:

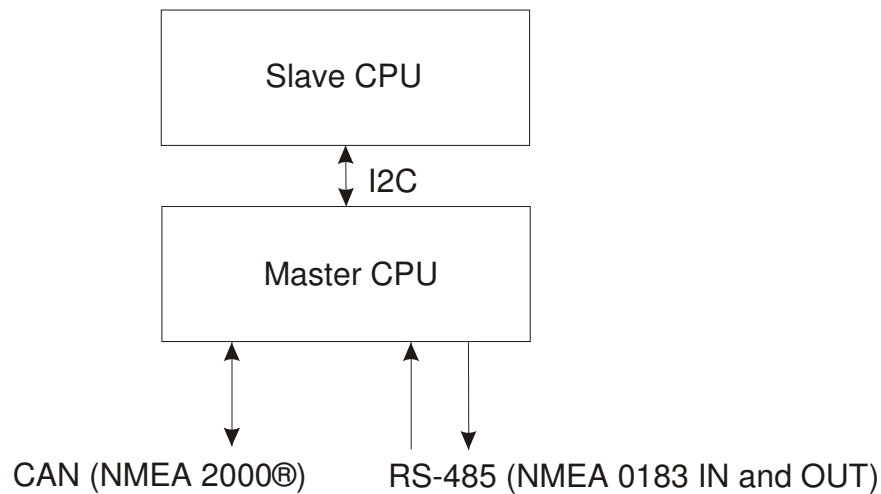
- Communication with host via either a full-duplex asynchronous serial RS-485 interface, or an NMEA 2000® Level B certified network interface based on CAN bus technology
- Perform firmware updates for multiple microcontrollers in the same product through a common command interface
- Firmware updates may be performed over a live NMEA 2000® network with no need to disable part of the network or reduce it to a special service state
- The bootloader program in each microcontroller is separate and distinct from the application program, and located in a physically separate region in flash memory, to prevent its inadvertent erasure or corruption
- Supports encryption of firmware being transmitted to the unit to provide protection of intellectual property

2. Related Documents

- *NMEA 0183 Standard for Interfacing Marine Electronic Devices, v3.01 (National Marine Electronics Association)*
- *NMEA 2000® Standard for Serial-Data Networking of Marine Electronic Devices, version 1.200 (National Marine Electronics Association)*

3. Overview

This overview will use the PB200 WeatherStation as an example. The PB200 incorporates two microcontrollers, each containing flash memory. A block diagram is shown below:



The two microcontrollers are referred to as the Master CPU and the Slave CPU.

The flash memory of each of the microcontrollers is partitioned into two sections: a bootloader section, and an application section. In each microcontroller the bootloader and application sections are physically separate regions of memory, containing separate and distinct programs.

The primary purpose of the bootloader program in each microcontroller is to facilitate performing field upgrades to the firmware stored in the application section of flash memory for that microcontroller.

The bootloader program itself is not field upgradeable. In the event the bootloader in a given unit becomes damaged or must be upgraded, the unit must be sent to the Airmar factory for reprogramming.

An external host computer or control/display unit sends commands and data to the Master CPU bootloader via either the RS-485 (NMEA 0183) or the CAN (NMEA 2000®) interface. Commands and data are further transferred from the Master CPU to the Slave CPU bootloader via an I2C (inter-integrated circuit) interface.

A common host command interface is used to perform a common set of operations on both microcontrollers. These operations may be performed using commands sent over the serial RS-485 interface, or PGNs sent over the CAN interface.

At power up or system reset, a Startup Monitor in the Master CPU is used to direct control to the serial bootloader, the CAN bootloader, or the application program.

4. Boot Sequence and Startup Monitor

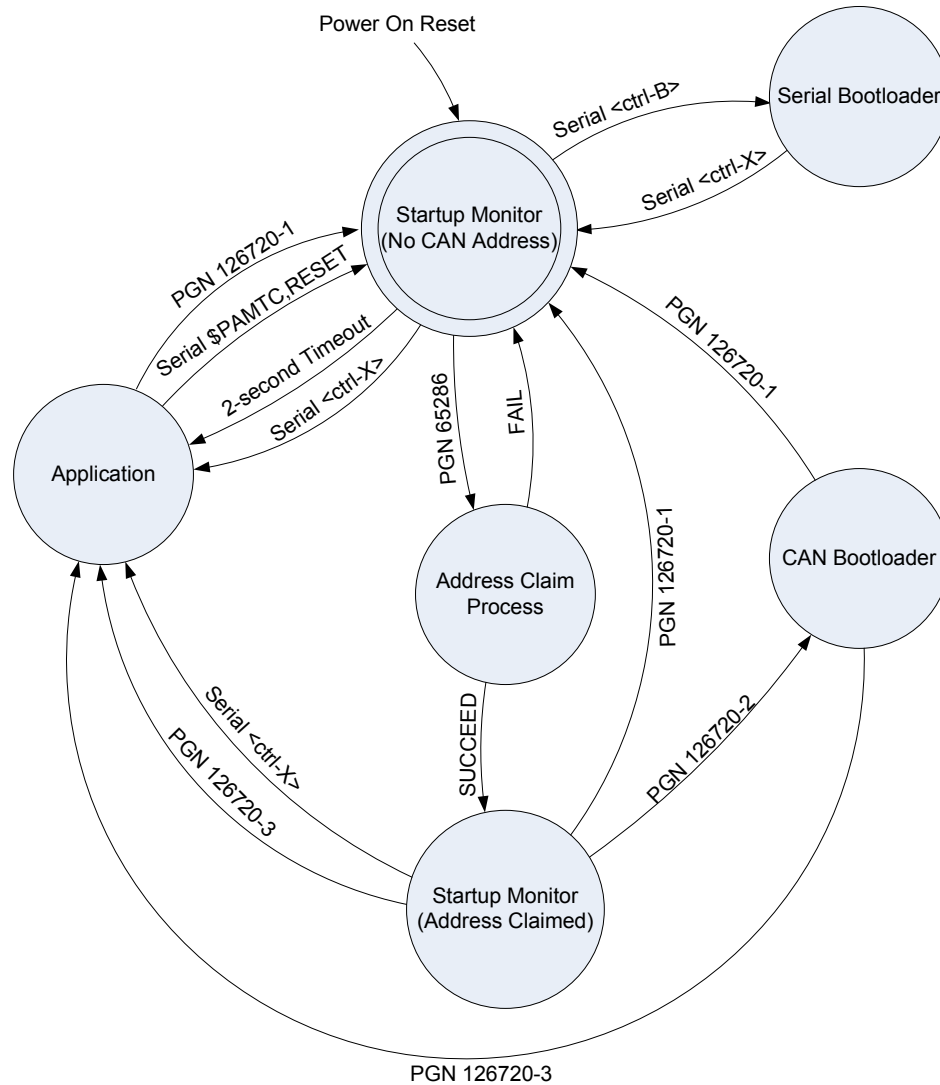
On power up or after a reset, the following sequence of steps is performed:

- 4.1. Execution in the Master CPU begins in the bootloader sector in flash memory. I/O ports are initialized based on the anticipated requirements of the application program. The serial UART associated with the RS-485 interface is initialized for 4800 baud, 8 data bits, no parity, and 1 stop bit. The CAN controller is initialized for a bit rate of 250 kbps.
- 4.2. A simple integrity check is performed on the application sector in flash memory to determine if a valid executable program is present.
- 4.3. The Startup Monitor is then executed. The Startup Monitor listens on both the RS-485 and CAN interfaces for certain commands to be received. These commands are described below.
- 4.4. If step 4.2. determined that the application sector of flash memory contains a valid executable program, then the Startup Monitor will continue for a 2-second interval, after which if no recognized command was received, then the Startup Monitor is terminated and the application program is launched. Otherwise, with no valid application program to launch, the unit will remain in the Startup Monitor indefinitely until a recognized command is received.
- 4.5. If the Startup Monitor receives an ASCII <ctrl-B> (0x02) on the RS-485 interface, then the unit shall immediately terminate the Startup Monitor, and launch the serial bootloader program. The bootloader will transmit (on the RS-485 interface) the bootloader sign-on message, followed by the bootloader prompt "01>".
- 4.6. If the Startup Monitor receives an ASCII <ctrl-X> (0x18) on the RS-485 interface, then the unit shall immediately terminate the Startup Monitor, and launch the application program stored in its flash memory. If step 4.2. determined that no valid application program exists, then the unit shall ignore the received <ctrl-X>, and remain in the Startup Monitor.
- 4.7. If the Startup Monitor receives any characters other than <ctrl-B> or <ctrl-X> on the RS-485 interface, these characters are ignored.
- 4.8. If the Startup Monitor (pre-address claim) receives the Airmar proprietary Boot State Request PGN 65286 via the CAN interface, then the unit shall claim a CAN address according to the NMEA 2000® address claim procedure, then transmit the proprietary Boot State Acknowledgment PGN 65285. The unit shall remain in the Startup Monitor context, but now having claimed an address, can recognize and respond to further NMEA 2000® requests received on the bus.

- 4.9. If the Startup Monitor (pre-address claim) receives any PGNs other than the Airmar proprietary Boot State Request PGN 65286, these PGNs shall be ignored, as though the device were off-bus.
- 4.10. If the Startup Monitor (post-address claim) receives either an ISO request (PGN 59904) or a complex request (PGN 126208) for one of the following standard NMEA 2000® PGNs, it shall provide the appropriate response as required by the NMEA 2000® standard:
- 60928 Address Claim
 - 126464 PGN List - Transmit and Received PGNs Group Function
 - 126996 Product Information
- 4.11. If the Startup Monitor (post-address claim) receives a complex request (PGN 126208) for one of the following Airmar proprietary PGNs, it shall provide the appropriate response:
- 126720-4 Startup Monitor Timeout
 - 126720-5 Bootloader Version
 - 126720-6 Bootloader Status
- 4.12. If the Startup Monitor (post-address claim) receives one of the following Airmar proprietary PGNs, it shall respond with the appropriate action.
- 65286 Boot State Request
 - 126720-1 Master Reset
 - 126720-2 Launch bootloader
 - 126720-3 Launch application

The power-up boot sequence is illustrated by the state diagram in Figure 1.

Figure 1. Boot sequence state diagram



5. CAN Bootloader

The Airmar NMEA 2000® / NMEA 0183 products have a single CAN interface, and are expected to be certified as NMEA 2000®-compliant Class 1 Level B devices. The bootloader transmits and receives standard and proprietary NMEA 2000® Parameter Groups via this interface. Transmitted PGNs (Parameter Group Numbers) are described in section 5.2. Received PGNs are described in section 5.3.

While the CAN bootloader is executing, the serial interface (RS-485/NMEA 0183) on the device is non-operational.

5.1. Terminology

The following terms related to the NMEA 2000® protocol appear on the detailed PGN pages to follow.

PGN: Parameter Group Number. NMEA 2000® messages are organized into logical collections of data called Parameter Groups. The NMEA assigns each Parameter Group a unique Parameter Group Number, or PGN.

Single Frame: Yes = the Parameter Group has been designed to fit within the 8 bytes of a single CAN frame. No = the Parameter Group is larger than 8 bytes, so must be transmitted in multiple CAN frames using either NMEA 2000® Fast Packets, or (TBD) ISO 11783 multi-packets, using the Data Transfer Transport Protocol (PGNs 60160 and 60416).

Priority: NMEA 2000® parameter groups are each assigned a priority value between 0 and 7, which is used to arbitrate between two devices attempting to transmit simultaneously. 0 is the highest priority, and 7 is the lowest. The Priority field at the top of each of the detailed specification pages shows three values separated by slashes (e.g. 6/Y/Y). The first value is the default priority, between 0 and 7, of the parameter group as defined in the NMEA 2000® Standard. The second value is "Y" if the priority for this parameter group may be changed using the Command Group Function PGN 126208, or "N" if the priority may not be changed. The third value is "Y" if the priority is saved in nonvolatile EEPROM memory within the device and maintained upon cycling power, or "N" if the priority reverts to the default value upon cycling power.

Update Rate: Some NMEA 2000® parameter groups are transmitted periodically by the device. The Update Rate field at the top of each of the detailed specification pages shows three values separated by slashes (e.g. 100 ms/Y/Y). The first value is the default update rate for the parameter group, as defined in the NMEA 2000® Standard. If the default behavior for the parameter group is not to transmit periodically, then this first value will be 0 (zero). The second value is "Y" if the update rate for this parameter group may be changed using the

Transmission Interval field in the Request Group Function PGN 126208, or "N" if the update rate may not be changed. The third value is "Y" if the update rate is saved in nonvolatile EEPROM memory within the device and maintained upon cycling power, or "N" if the update rate reverts to the default value upon cycling power.

Destination: Global or Address. A Global Destination means that the parameter group was designed to be broadcast for use by all receiving devices on the bus. If Destination = Address, this means the parameter group was designed to be transmitted to a specific device on the bus. In this case, the address of the destination device must be provided within the PGN.

Query Support: If the parameter group supports queries, then the unit will transmit the parameter group if requested by another device on the bus. The request must be made using either the ISO Request PGN 59904, or the Complex Request Group PGN 126208.

Access Level: The bootloader contains a security mechanism whereby certain PGNs require an access sequence to be performed in order to enable the functionality in the PGN. If the indicated Access Level is 0, then the PGN is fully functional without requiring an access sequence to be performed. If the indicated Access Level is nonzero, then it is necessary to perform an access sequence using the proprietary transmitted PGN 065287, Access Level, before the functionality of the PGN will be enabled.

Request Parameter: A "Yes" value specifies that the given field may be used by the Request Group Function PGN 126208 as a qualifier to request specific data according to the contents of the field.

DD: Data Dictionary. The Data Dictionary number defines the contents of the parameter group field. The designators in this column are described in detail in the NMEA 2000® Standard, Appendix B.2.

DF: Data Format. The Data Format number defines the format of the data in the parameter group field. The designators in this column are described in detail in the NMEA 2000® Standard, Appendix B.3.

Type: The Type designators are described in detail in the NMEA 2000® Standard, Appendix B.4.

Resolution: The resolution indicates the smallest increment of the value in the given field, in the units specified by the data format.

Value: This indicates the range of allowable values the field may contain.

5.2. Transmitted NMEA 2000® PGNs

The NMEA 2000® PGNs transmitted by the bootloader are summarized in Table 1. The contexts in which each of the PGNs may be transmitted are indicated in the three rightmost columns in the table. Each of the transmitted PGNs in Table 1 is described in detail on the subsequent pages.

Table 1: NMEA 2000® Transmitted PGNs

<u>PGN</u>	<u>Description</u>	<u>SM</u>	<u>SM(AC)</u>	<u>BL</u>
59392	ISO Acknowledgement	-	✓	✓
60928	ISO Address Claim	-	✓	✓
65285	Proprietary: Boot State Acknowledgment	-	✓	✓
65287	Proprietary: Access Level	-	✓	✓
126208	Acknowledge Group Function	-	✓	✓
126464	PGN List – Transmit and Received PGNs Group Function	-	✓	✓
126720	Proprietary:			
	-4: Startup Monitor Timeout	-	✓	-
	-5: Bootloader Version	-	-	✓
	-6: Bootloader Status	-	-	✓
126996	Product Information	-	✓	✓

Note:

SM = Startup Monitor (prior to address claim)

SM(AC) = Startup Monitor (after address claim)

BL = Bootloader

PGN 059392 (0x0E800)

ISO Acknowledgment

Single Frame: Yes	Priority: 6/N/N	Update Rate: 0/N/N	Destination: Address	Query Support: Yes	Access Level: 0
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Field #	Name	Request Param.	DD	DF	Type	Resolution	Value	Remark
1	Control Byte	No	DD177	DF52	8 bits	1	0x00 = Positive Acknowledgment 0x01 = Negative Acknowledgment 0x02 = PGN supported but access denied 0x03 to 0xFF = Reserved	
2	Group Function Value	No	DD178	DF52	8 bits	1	Variable	Group Function of PGN being acknowledged. This field is not used if the PGN being acknowledged or declined is not a group function PGN.
3	Reserved bits	No	DD001	DF52	24 bits	1	0xFFFFFFFF	All bits set to logic "1"
4	PGN of Requested Information	No	DD009	DF52	24 bits	1	Variable	24-bit Parameter Group Number (PGN) expressed in binary, LSB is transmitted first.

Notes

This PGN is transmitted as a handshake mechanism in response to an ISO Request PGN 59904.

PGN 060928 (0x0EE00)

ISO Address Claim

Single Frame: Yes	Priority: 6/N/N	Update Rate: 0/N/N	Destination: Address	Query Support: Yes	Access Level: 0
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Field #	Name	Request Param.	DD	DF	Type	Resolution	Value	Remark
1	Unique Number (ISO Identity Number)	No	DD173	DF52	21 bits	1	0x000000 to 0x1FFFFFFF	This value is programmed during the factory calibration and test procedure to be a unique value from one individual unit to the next. The number's uniqueness is dependent upon the Manufacturer Code (field 2), Device Function (field 5), and Device Class (field 7). The value is stored in EEPROM, but is not changeable in the field.
2	Manufacturer Code	Yes	DD172	DF52	11 bits	1	135 (decimal)	Manufacturer Code for Airmar Technology.
3	Device Instance Lower (ISO ECU Instance)	Yes	DD201	DF52	3 bits	1	Default value = 0	Value stored in EEPROM.
4	Device Instance Upper (ISO Function Instance)	Yes	DD174	DF52	5 bits	1	Default value = 0	Value stored in EEPROM.
5	Device Function (ISO Function)	Yes	DD171	DF52	8 bits	1	180 (decimal)	NMEA 2000® Function Code for "Weather Instruments".
6	Reserved (Dominant bit)	No	DD175	DF52	1 bit	1	0	Dominant bit = 0
7	Device Class	Yes	DD170	DF52	7 bits	1	80 (decimal)	NMEA 2000® Class Code for "Instrumentation/general systems".
8	System Instance (ISO Device Class Instance)	Yes	DD169	DF52	4 bits	1	Default value = 0	Value stored in EEPROM.
9	Industry Group	Yes	DD168	DF52	3 bits	1	4 (decimal)	Industry Group for "Marine Industry".
10	Reserved (ISO Self Configurable)	No	DD001	DF52	1 bit	1	0x1	All bits set to logic "1"

Notes

PGN 60928 is a network management message used to claim a network address, reply to devices requesting the claimed address, and to respond to requests for device information (NAME). The 64 bits comprised by the ten fields in this parameter group constitute the ISO 11783-5 NAME entity.

The contents of fields 3, 4, and 8 are stored in EEPROM memory and may be changed using PGN 126208, Command Group Function.

PGN 065285 (0x0FF05)

Proprietary: Boot State Acknowledgment

Single Frame: Yes Priority: 7/N/N Update Rate: 0/N/N Destination: Global Query Support: No Access Level: 0

Field #	Name	Request Param.	DD	DF	Type	Resolution	Value	Remark
1	Manufacturer Code	No	DD172	DF52	11 bits	1	135 (decimal)	Manufacturer Code for Airmar Technology.
2	Reserved bits	No	DD001	DF52	2 bits	1	0x3	All bits set to logic "1"
3	Industry Group	No	DD168	DF52	3 bits	1	4 (decimal)	Industry Group for "Marine Industry".
4	Boot State	No	N/A	DF52	3 bits	1	000 = in Startup Monitor 001 = running Bootloader 010 = running Application 011 to 110 = reserved 111 = Data Not Available	
5	Reserved bits	No	DD001	DF52	45 bits	1	0x1FFFFFFFFFFFF	All bits set to logic "1"

Notes

The Boot State Acknowledgment proprietary PGN is transmitted in response to the received proprietary PGN 65286, Global Boot State Request. This PGN is transmitted from any of the following contexts: Startup Monitor, CAN Bootloader, or Application.

PGN 065287 (0x0FF07)

Proprietary: Access Level

Single Frame: Yes	Priority: 7/N/N	Update Rate: 0/N/N	Destination: Global	Query Support: Yes	Access Level: 0
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Field #	Name	Request Param.	DD	DF	Type	Resolution	Value	Remark
1	Manufacturer Code	Yes	DD172	DF52	11 bits	1	135 (decimal)	Manufacturer Code for Airmar Technology.
2	Reserved bits	No	DD001	DF52	2 bits	1	0x3	All bits set to logic "1"
3	Industry Group	Yes	DD168	DF52	3 bits	1	4 (decimal)	Industry Group for "Marine Industry".
4	Format Code	Yes	DD005	DF53	uint8	1 bit	1	1 = Format Code 1
5	Access Level	No	N/A	DF52	3 bits	1	0x0 = locked 0x1 = unlocked, level 1 0x2 = unlocked, level 2 0x3 = unlocked, level 3 0x4 = reserved 0x5 = unlocked, level 5 0x6 = set level 1 password 0x7 = reserved	When this PGN is transmitted, this field indicates the current Access Level of the device. This field may be changed using the Command Group Function PGN 126208 (see text below).
6	Reserved bits	No	DD001	DF52	5 bits	1	0x1F	All bits set to logic "1"
7	Access Seed/Key	No	N/A	DF55	uint32	1 bit	0 to 4,294,967,292	When this PGN is transmitted, this field contains a Seed that may be used in a subsequent Unlock operation. This field is also used to provide a Key to the device when performing an Unlock operation, using the Command Group Function PGN 126208 (see text below).

Notes

The Access Level proprietary PGN is a security mechanism that provides the device with a method to limit access to certain functions to devices that understand the proper unlock sequence. Certain operations will not be possible unless the Access Level is first increased to the required value. A nonzero Access Level remains in effect until the unit powers down, or 15 minutes has elapsed, whichever comes first, after which the Access Level returns to zero. The Access Level is not stored in EEPROM.

This PGN is transmitted in response to a PGN 126208 Request Group Function message. Fields 1 and 3 must each be fully specified in the request in order for this PGN to be transmitted.

This PGN does not respond to ISO Requests (PGN 59904).

The NMEA 2000® Bootloader and Startup Monitor recognize certain PGNs that use Access Levels 0, 1, and 5. The method for enabling these Access Levels is described below. When PGN 65287 is transmitted, field 5 contains the current Access Level of the device, and field 7 contains a randomly-generated Seed that may be used in a subsequent Unlock operation.

To change the Access Level, use the Command Group Function PGN 126208, specifying the individual fields in PGN 65287 as follows:

To change the Access Level to 0 (i.e. to lock the device after having been previously unlocked):

Use the Command Group Function PGN 126208, specifying the individual fields in PGN 65287 as follows. (Note that it is not required to specify field 7.)

PGN 65287	
Field	Value
1	135 (decimal)
3	4 (decimal)
4	1
5	0

The device will respond with the Acknowledge Group Function PGN 126208, indicating success or failure according to the error codes provided in the parameter fields.

To change the Access Level to 1:

Use the Command Group Function PGN 126208, specifying the individual fields in PGN 65287 as follows.

Field 7 serves as a simple password to allow setting the Access Level to 1. For this Access Level, it is not necessary to obtain a Seed or calculate a Key.

PGN 65287	
Field	Value
1	135 (decimal)
3	4 (decimal)
4	1
5	1
7	0x12345678

The device will respond with the Acknowledge Group Function PGN 126208, indicating success or failure according to the error codes provided in the parameter fields.

To change the Access Level to 5:

1. Use the Request Group Function PGN 126208, specifying PGN 65287, to obtain a Seed.
2. The device will respond with PGN 65287, with the current Access Level in field 5, and with a random Seed in field 7.

3. Using the Seed provided in step 2, calculate a Key according to the following algorithm, where the mask argument is the value 0x64C257DA, and the number of iterations is 67.

```
uint32 calculate_level_2_key (uint32 seed, uint32 mask, uint8 iterations)
{
    uint8 i;
    uint32 shift_reg;
    uint32 temp;

    shift_reg = seed ^ mask;

    if (shift_reg == 0)                /* input must be nonzero */
        shift_reg = 0xFFFFFFFF;

    for (i=0; i<iterations; i++) {
        temp = shift_reg >> 31;
        temp ^= shift_reg >> 6;
        temp ^= shift_reg >> 5;
        temp ^= shift_reg >> 1;
        temp &= 0x00000001;
        shift_reg = (temp << 31) | (shift_reg >> 1);
    }
    return shift_reg;
}
```

Note that the above algorithm along with the given mask and iterations arguments is proprietary and confidential information specific to Airmar Technology Corporation, and is not to be shared with any third party.

4. Use the Command Group Function PGN 126208, specifying the individual fields in PGN 65287 as follows:

PGN 65287	
Field	Value
1	135 (decimal)
3	4 (decimal)
4	1
5	The desired Access Level
7	The calculated Key

5. The device will respond with the Acknowledge Group Function PGN 126208, indicating success or failure according to the error codes provided in the parameter fields.

After executing the above sequence to change the Access Level, you may optionally request this PGN using the Request Group Function PGN 126208 to confirm that the Access Level was in fact changed.

Note that increasing the Access Level does not in and of itself allow global access to the secure functions of the bootloader. The Access Level is changed only for subsequent commands and requests originating from the device with the source address originally used to specify the Access Level.

NMEA 2000® TRANSMITTED PARAMETER GROUP

PGN 126208 (0x1ED00)

Acknowledge Group Function

Single Frame: No	Priority: 6/N/N	Update Rate: 0/N/N	Destination: Address	Query Support: Yes	Access Level: 0
------------------	-----------------	--------------------	----------------------	--------------------	-----------------

Field #	Name	Request Param.	DD	DF	Type	Resolution	Value	Remark
1	Group Function Code	No	DD144	DF53	uint8	1 bit	0 = Request Message 1 = Command Message 2 = Acknowledge Message 3 = Read Fields 4 = Read Fields Reply 5 = Write Fields 6 = Write Fields Reply	The unit transmits only the Acknowledge Message, so this field will always contain the value 2.
2	Requested or Commanded PGN # being acknowledged	No	DD009	DF52	24 bits	1	Variable	24-bit Parameter Group Number (PGN) expressed in binary, LSB is transmitted first.
3	PGN Error Code	No	DD037	DF52	4 bits	1	0x0 = No Error/Acknowledge 0x1 = PGN not supported 0x2 = PGN temporarily not available 0x3 = Access denied 0x4 = Request is not supported	
4	Transmission Interval/Priority error code	No	DD139	DF52	4 bits	1	0x0 = No Error/Acknowledge 0x1 = Transmit Interval/Priority not supported 0x2 = Transmit interval is less than measurement/calculation interval 0x3 = Access denied 0x4 = Request is not supported	
5	Number of Requested or Commanded Parameters	No	DD006	DF53	uint8	1 bit		Value 0xFF indicates no fields follow

Field #	Name	Request Param.	DD	DF	Type	Resolution	Value	Remark
6	First parameter error code	No	DD141	DF52	4 bits	1	0x0 = No Error/Acknowledge 0x1 = Invalid request or command parameter field 0x2 = Temporarily unable to comply 0x3 = Request or command parameter out-of-range 0x4 = Access denied 0x5 = Request or Command Group Function not supported	
7	Variable number of fields, Field 6 repeated	No	DD141	DF52	4 bits	1	0x0 = No Error/Acknowledge 0x1 = Invalid request or command parameter field 0x2 = Temporarily unable to comply 0x3 = Request or command parameter out-of-range 0x4 = Access denied 0x5 = Request or Command Group Function not supported	

Notes

The Acknowledgment Reply is transmitted in response to a PGN 126208 Request or Command Group Function message. In response to the Request message, the Acknowledge is only transmitted for a request that cannot be complied with.

See also related PGNs:

- the received Request Group Function PGN 126208
- the received Command Group Function PGN 126208

PGN 126464 (0x1EE00)

PGN List - Transmit/Received PGNs Group

Single Frame: No	Priority: 6/N/N	Update Rate: 0/N/N	Destination: Address	Query Support: Yes	Access Level: 0
------------------	-----------------	--------------------	----------------------	--------------------	-----------------

Field #	Name	Request Param.	DD	DF	Type	Resolution	Value	Remark
1	Group Function Code	Yes	DD145	DF53	uint8	1 bit	0 = Transmit PGN List Message 1 = Received PGN List Message	
2	First PGN supported	No	DD009	DF52	24 bits	1	Variable	24-bit Parameter Group Number (PGN) expressed in binary, LSB is transmitted first.
3	Variable Number of fields, Field 2 repeated	No	DD009	DF52	24 bits	1	Variable	24-bit Parameter Group Number (PGN) expressed in binary, LSB is transmitted first.

Notes

This message provides a list of the transmitted or received PGNs that are supported by the bootloader. It is transmitted only upon request. If it is requested using the ISO Request PGN 59904, or with the Request Group Function PGN 126208 without specifying the Group Function Code in field 1, then this PGN will be transmitted twice, once each for the transmitted and received lists.

PGN 126720 (0x1EF00)

Addressable Multi-Frame Proprietary

Single Frame: No	Priority: 7/Y/N	Update Rate: 0/N/N	Destination: Address	Query Support: Yes	Access Level: 0
------------------	-----------------	--------------------	----------------------	--------------------	-----------------

Field #	Name	Request Param.	DD	DF	Type	Resolution	Value	Remark
1	Manufacturer Code	Yes	DD172	DF52	11 bits	1	135 (decimal)	Manufacturer Code for Airmar Technology.
2	Reserved bits	No	DD001	DF52	2 bits	1	0x3	All bits set to logic "1"
3	Industry Group	Yes	DD168	DF52	3 bits	1	4 (decimal)	Industry Group for "Marine Industry".
4	Proprietary ID (PID)	Yes	DD005	DF53	uint8	1 bit	0 to 252	Identifier specifying how to interpret the remaining fields in this parameter group.
5+	(PID-dependent)	No	DD000	DF00	undef.	undefined	undefined	The use of the remaining bytes and the number of CAN frames required varies according to the PID (field 4).

Notes

This PGN is transmitted in response to a PGN 126208 Request Group Function message. Fields 1, 3, and 4 must each be fully specified in the request in order for this PGN to be transmitted.

This PGN does not respond to ISO Requests (PGN 59904).

The use of this PGN with the following Proprietary ID's (field 4) is described on the subsequent pages:

- 4: Startup Monitor Timeout
- 5: Bootloader Version
- 6: Bootloader Status

PGN 126720-4 (0x1EF00)

Proprietary: Startup Monitor Timeout

Single Frame: No	Priority: 7/Y/N	Update Rate: 0/N/N	Destination: Address	Query Support: Yes	Access Level: 0
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Field #	Name	Request Param.	DD	DF	Type	Resolution	Value	Remark
1	Manufacturer Code	Yes	DD172	DF52	11 bits	1	135 (decimal)	Manufacturer Code for Airmar Technology.
2	Reserved bits	No	DD001	DF52	2 bits	1	0x3	All bits set to logic "1"
3	Industry Group	Yes	DD168	DF52	3 bits	1	4 (decimal)	Industry Group for "Marine Industry".
4	Proprietary ID (PID)	Yes	DD005	DF53	uint8	1 bit	4 (decimal)	4 = Startup Monitor Timeout
5	Time Remaining	No	DD211	DF81	uint16	0.1 sec	0 to 65532 = 0 to 6553.2 seconds 65533 = Reserved 65534 = Out of Range (infinite timeout) 65535 = Data Not Available or Do Not Change	This value indicates the amount of time remaining in the Startup Monitor before the application program is launched. This field may be initialized to any value from 0 to 65532, or 65534, using the Command Group Function PGN 126208. If there is no valid application program in flash memory, then this field will contain 65534.
6	Reserved bits	No	DD001	DF52	8 bits	1	0xFF	All bits set to logic "1"

Notes

This PGN is only recognized when the device is operating in the Startup Monitor context.

This PGN is transmitted in response to a PGN 126208 Request Group Function message. Fields 1, 3, and 4 must each be fully specified in the request in order for this PGN to be transmitted.

This PGN does not respond to ISO Requests (PGN 59904).

(continued on next page)

To change the Time Remaining (field 5), use the Command Group Function PGN 126208, specifying the individual fields in PGN 126720-4 as follows.

PGN 126720	
Field	Value
1	135 (decimal)
3	4 (decimal)
4	4 (decimal)
5	0-65532 = time, from 0 to 6553.2 seconds 65534 = set timeout to infinite

This PGN may only be requested or commanded when the device is in the Startup Monitor context. Note that if the timeout value is nearly zero, meaning the context is on the verge of being switched to the Application, then a request or command for this PGN might be ignored.

PGN 126720-5 (0x1EF00)

Proprietary: Bootloader Version

Single Frame: No	Priority: 7/Y/N	Update Rate: 0/N/N	Destination: Address	Query Support: Yes	Access Level: 0
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Field #	Name	Request Param.	DD	DF	Type	Resolution	Value	Remark
1	Manufacturer Code	Yes	DD172	DF52	11 bits	1	135 (decimal)	Manufacturer Code for Airmar Technology.
2	Reserved bits	No	DD001	DF52	2 bits	1	0x3	All bits set to logic "1"
3	Industry Group	Yes	DD168	DF52	3 bits	1	4 (decimal)	Industry Group for "Marine Industry".
4	Proprietary ID (PID)	Yes	DD005	DF53	uint8	1 bit	5 (decimal)	5 = Bootloader Version
5	Product Code	No	DD007	DF54	uint16	1 bit	4 (decimal) for the PB200 Codes for other products TBD.	This field provides the same information that the Serial Bootloader provides in its reply to the C command. Note that this field is not to be confused with the NMEA 2000® Manufacturer's Product Code in field 2 of the Product Information PGN 126996.
6	Processor Code	No	DD005	DF53	uint8	1 bit	For the PB200 WeatherStation: 0 = reserved 1 = Master CPU 2 = Slave CPU 3 to 255 = reserved Other products TBD.	This field indicates the CPU context of the following fields. It provides the same information that the Serial Bootloader provides in its command prompt. The received PGN 126720-10, command 6, may be used to switch the context to a different processor code.
7	Bootloader Version	No	DD007	DF54	uint16	1 bit	0 to 65532	This field provides the same information that the Serial Bootloader provides in its reply to the B command. There is an implied decimal point located 3 decimal digits from the right. For example, a bootloader with a version number of 1.003 would be reported as 1003 decimal, or 0x03EB.

Field #	Name	Request Param.	DD	DF	Type	Resolution	Value	Remark
8	Application Version	No	DD007	DF54	uint16	1 bit	0 to 65532	This field provides the same information that the Serial Bootloader provides in its reply to the A command. There is an implied decimal point located 3 decimal digits from the right. For example, application firmware with a version number of 1.002 would be reported as 1002 decimal, or 0x03EA.

Notes

This PGN is transmitted in response to a PGN 126208 Request Group Function message. Fields 1, 3, and 4 must each be fully specified in the request in order for this PGN to be transmitted.

This PGN does not respond to ISO Requests (PGN 59904).

PGN 126720-6 (0x1EF00)

Proprietary: Bootloader Status

Single Frame: No	Priority: 7/Y/N	Update Rate: 0/N/N	Destination: Address	Query Support: Yes	Access Level: 0
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Field #	Name	Request Param.	DD	DF	Type	Resolution	Value	Remark
1	Manufacturer Code	Yes	DD172	DF52	11 bits	1	135 (decimal)	Manufacturer Code for Airmar Technology.
2	Reserved bits	No	DD001	DF52	2 bits	1	0x3	All bits set to logic "1"
3	Industry Group	Yes	DD168	DF52	3 bits	1	4 (decimal)	Industry Group for "Marine Industry".
4	Proprietary ID (PID)	Yes	DD005	DF53	uint8	1 bit	6 (decimal)	6 = Bootloader Status
5	SID (Sequence ID)	No	DD056	DF53	uint8	1 bit	0 to 252	Unit-less number. If this PGN is being transmitted in response to a received PGN 126720-10 or 126720-11, then this field will contain the same value as provided in the SID field in the received PGN. If this PGN is being transmitted in response to a Request Group Function PGN 126208, then this field will contain the value 255.
6	Processor Code	No	DD005	DF53	uint8	1 bit	For the PB200 WeatherStation: 0 = reserved 1 = Master CPU 2 = Slave CPU 3 to 255 = reserved Other products TBD.	This field indicates the CPU context of the following fields. It provides the same information that the Serial Bootloader provides in its command prompt. The received PGN 126720-10, command 6, may be used to switch the context to a different processor code.
7	Programming Mode	No	N/A	DF52	3 bits	1	0 = Locked, can't program 1 = Unlocked 2 = Unlocked, Program mode 3 = Unlocked, Verify mode 4 to 7 = reserved	The state of this field must be changed using the received PGN 126720-10.

Field #	Name	Request Param.	DD	DF	Type	Resolution	Value	Remark
8	Status of last operation	No	N/A	DF52	5 bits	1	0 to 31	0 = Operation successful; no error 1 = Program or Verify operation completed and successful 2 = Program or Verify operation completed unsuccessfully 3 = Reserved 4 = Reserved 5 = Input buffer overflow 6 = Erase error 7 = Checksum error 8 = Header error 9 = S-record error 10 = Sequence (SID) error 11 = Flash Write or Compare Error 12 = Flash Write or Compare No Error 13 to 29 = Reserved 30 = Other error 31 = Data Not Available

Notes

This PGN is transmitted in response to a PGN 126208 Request Group Function message. Fields 1, 3, and 4 must each be fully specified in the request in order for this PGN to be transmitted.

This PGN is also transmitted in response to a received PGN 126720-10 Bootloader Command message, or a received PGN 126720-11 Bootloader Data message.

This PGN does not respond to ISO Requests (PGN 59904).

Although this is a fast-packet message, the version with this PID fits into a single CAN frame.

NMEA 2000® TRANSMITTED PARAMETER GROUP

PGN 126996 (0x1F014)

Product Information

Single Frame: No Priority: 6/N/N Update Rate: 0/N/N Destination: Global Query Support: Yes Access Level: 0

Field #	Name	Request Param.	DD	DF	Type	Resolution	Value	Remark
1	NMEA 2000® Database Version	No	DD007	DF54	uint16	1 bit	1200 (decimal), corresponding to NMEA 2000® database version 1.200.	
2	NMEA Manufacturer's Product Code	Yes	DD007	DF54	uint16	1 bit	Depends on product.	This product code is uniquely assigned by the NMEA for exclusive use in the specific product.
3	Manufacturer's Model ID	No	DD192	DF63	char8(n)	1 char	Fixed length, 32-character ASCII string, e.g.: "PB200 WeatherStation"	Enclosing quotes not included
4	Manufacturer's Software Version Code	No	DD192	DF63	char8(n)	1 char	Fixed length, 32-character ASCII string. For example: "1.000,1.000,1.000,1.000"	Enclosing quotes not included. The four values separated by commas represent, in order: • Bootloader 1 firmware version • Application 1 firmware version • Bootloader 2 firmware version • Application 2 firmware version
5	Manufacturer's Model Version	No	DD192	DF63	char8(n)	1 char	Fixed length, 32-character ASCII string.	This value is stored in EEPROM memory, but is not changeable by the user.
6	Manufacturer's Model Serial Code	No	DD192	DF63	char8(n)	1 char	Fixed length, 32-character ASCII string, representing the product serial number.	This value is stored in EEPROM memory, but is not changeable by the user.
7	NMEA 2000® Certification Level	No	DD226	DF53	uint8	1 bit	1	0 = Level A 1 = Level B

Field #	Name	Request Param.	DD	DF	Type	Resolution	Value	Remark
8	Load Equivalency	No	DD257	DF53	uint8	1 bit	TBD	Unit-less number. Defined in the NMEA 2000® Standard's Main Document in section 2.4.7 Interface Power.

Notes

5.3. Received NMEA 2000® PGNs

The NMEA 2000® PGNs recognized by the bootloader are summarized in Table 2. The contexts in which each of the PGNs is recognized are indicated in the three rightmost columns in the table. Each of the PGNs in Table 2 is described in detail on the subsequent pages.

Table 2: NMEA 2000® Received PGNs

<u>PGN</u>	<u>Description</u>	<u>SM</u>	<u>SM(AC)</u>	<u>BL</u>
59904	ISO Request	-	✓	✓
60928	ISO Address Claim	-	✓	✓
65286	Proprietary: Boot State Request	✓	✓	✓
126208	Request Group Function	-	✓	✓
126208	Command Group Function	-	✓	✓
126720	Proprietary:			
	-1: Master Reset	-	✓	✓
	-2: Launch bootloader	-	✓	-
	-3: Launch application	-	✓	✓
	-10: Bootloader command	-	-	✓
	-11: Bootloader data	-	-	✓

Note:

SM = Startup Monitor (prior to address claim)
 SM(AC) = Startup Monitor (after address claim)
 BL = Bootloader

NMEA 2000® RECEIVED PARAMETER GROUP

PGN **059904** (0x0EA00)

ISO Request

Single Frame: Yes

Access Level: 0

Field #	Name	DD	DF	Type	Resolution	Value	Remark
1	PGN being requested	DD009	DF52	24 bits	1	Variable	24-bit Parameter Group Number (PGN) expressed in binary, LSB is transmitted first.

Notes

When this PGN is received by the bootloader, the unit will respond by transmitting either the PGN being requested, or the ISO Acknowledge PGN 59392.

See also the received Request Group Function PGN 126208.

NMEA 2000® RECEIVED PARAMETER GROUP

PGN 060928 (0x0EE00)

ISO Address Claim

Single Frame: Yes Access Level: 0

Field #	Name	DD	DF	Type	Resolution	Value	Remark
1	Unique Number (ISO Identity Number)	DD173	DF52	21 bits	1	0x000000 to 0x1FFFFFF	
2	Manufacturer Code	DD172	DF52	11 bits	1	Variable	
3	Device Instance Lower (ISO ECU Instance)	DD201	DF52	3 bits	1	0x0 to 0x7	
4	Device Instance Upper (ISO Function Instance)	DD174	DF52	5 bits	1	0x0 to 0x1F	
5	Device Function (ISO Function)	DD171	DF52	8 bits	1	Variable	
6	Reserved (Dominant bit)	DD175	DF52	1 bit	1	0	Dominant bit = 0
7	Device Class	DD170	DF52	7 bits	1	Variable	
8	System Instance (ISO Device Class Instance)	DD169	DF52	4 bits	1	0x0 to 0xF	
9	Industry Group	DD168	DF52	3 bits	1	Variable	
10	Reserved (ISO Self Configurable)	DD001	DF52	1 bit	1	0x1	All bits set to logic "1"

Notes

PGN 60928 is a network management message used to claim a network address, reply to devices requesting the claimed address, and to respond to requests for device information (NAME). The 64 bits comprised by the ten fields in this parameter group constitute the ISO 11783-5 NAME entity.

NMEA 2000® RECEIVED PARAMETER GROUP

PGN 065286 (0x0FF06)

Proprietary: Boot State Request

Single Frame: Yes Access Level: 0

Field #	Name	DD	DF	Type	Resolution	Value	Remark
1	Manufacturer Code	DD172	DF52	11 bits	1	135 (decimal)	Manufacturer Code for Airmar Technology.
2	Reserved bits	DD001	DF52	2 bits	1	0x3	All bits set to logic "1"
3	Industry Group	DD168	DF52	3 bits	1	4 (decimal)	Industry Group for "Marine Industry".
4	Reserved bits	DD001	DF52	48 bits	1	0xFFFFFFFFFFFF	All bits set to logic "1"

Notes

When this destination global PGN is received by the bootloader, the unit will respond by transmitting PGN 65285 (see page 11). It is intended that all Airmar devices connected to the NMEA 2000® bus will likewise also reply accordingly when receiving this PGN.

This PGN is recognized from any of the following contexts: Startup Monitor, CAN Bootloader, or Application. If the unit is in the Startup Monitor context when this PGN is received, and has not yet claimed a CAN address, then the unit will perform an address claim sequence prior to transmitting PGN 65285, after which the Startup Monitor timer will be re-initialized to 2 seconds.

NMEA 2000® RECEIVED PARAMETER GROUP

PGN 126208 (0x1ED00)

Request Group Function

Single Frame: No Access Level: 0

Field #	Name	DD	DF	Type	Resolution	Value	Remark
1	Group Function Code	DD144	DF53	uint8	1 bit	0 = Request Message 1 = Command Message 2 = Acknowledge Message 3 = Read Fields 4 = Read Fields Reply 5 = Write Fields 6 = Write Fields Reply	For the Request Group Function, this field = 0.
2	Requested PGN	DD009	DF52	24 bits	1	Variable	24-bit Parameter Group Number (PGN) expressed in binary, LSB is transmitted first.
3	Transmission interval	DD035	DF65	uint32	0.001 s	0 to $\approx 4.295 \times 10^6$ seconds 0x0000 0000 = turn off transmission 0xFFFF FFFE = restore default interval 0xFFFF FFFF = do not change interval	Time interval between data transmissions of the Requested PGN. Note that none of the PGNs transmitted by the bootloader may be set to transmit periodically.
4	Transmission interval offset	DD036	DF66	uint16	0.01 s	0 to 655.32 seconds. 0x0000 = transmit immediately 0xFFFF = do not change offset	Offset in transmit time from time of request command. If this PGN is received with 0xFFFF in this field and 0xFFFF FFFF in field 3, then the requested PGN will be transmitted without changing timing variables.
5	Number of Pairs of Request Parameters to follow	DD006	DF53	uint8	1 bit	0 to 252	
6	Field number of first requested parameter	DD005	DF53	uint8	1 bit	0 to 252	Field number within the Requested PGN.

Field #	Name	DD	DF	Type	Resolution	Value	Remark
7	Value of first requested parameter	DD000	DF00	undef.	undefined	undefined	Requested parameter size, type, and range of legal values are dependent on the Requested PGN and the specific request parameter field.
8	Variable number of fields, field 6 repeated	DD005	DF53	uint8	1 bit	0 to 252	
9	Variable number of fields, field 7 repeated	DD000	DF00	undef.	undefined	undefined	

Notes

When this PGN is received by the device, the unit will compare the specified Request Parameters in fields 6, 7 (and any following fields) against the contents of the actual fields in the requested PGN. The only fields that may be used as Request Parameters are those which indicate a "Yes" in the "Request Param." column in the detailed description for the specific transmitted PGN. If all specified parameters match the values in the corresponding fields, then the requested PGN will be transmitted. If any of the parameters do not match, then the unit will respond by transmitting the Acknowledge Group Function PGN 126208 with the appropriate error code(s) provided.

Normally, this PGN is also used to specify the Transmission Interval (Update Rate) between successive transmissions of a periodically transmitted PGN. However, in the bootloader, there are no PGNs that are transmitted periodically, so this feature is not used.

Refer to the detailed descriptions of the specific transmitted PGNs in this document for any exceptions or special cases with regard to using the Request Group Function PGN 126208 with each PGN.

All parameter value fields must be padded (with 1's) if necessary to ensure byte boundaries are adhered to.

When the Requested PGN field 2 (of PGN 126208) refers to a proprietary PGN, fields 1 and 3 (manufacturer code and industry group) of the proprietary PGN must be specified *first* in the list of field number and parameter value pairs in PGN 126208. All other field number and parameter value pairs may appear in any order in this message.

For proprietary PGNs described in this document, fields 1, 2, and 3 (manufacturer code, reserved bits, and industry group) of the proprietary PGN may optionally be combined into a single 16-bit field when specifying fields in PGN 126208. For Airmar proprietary PGNs, the 16-bit field is constructed as follows:

- bits 15-13 = binary 100 = decimal 4 (industry group for "Marine Industry")
- bits 12-11 = binary 11 = reserved bits
- bits 10-0 = binary 000 1000 0111 = decimal 135 (manufacturer code for Airmar Technology)

The concatenation of these 16 bits produces the binary value 1001 1000 1000 0111 = hexadecimal 0x9887. Therefore, instead of specifying the two fields, 11-bit field 1 and 3-bit field 3 for a proprietary PGN, it is permissible to specify a single 16 bit field 1, containing the value 0x9887, for the proprietary PGNs described in this document.

When doing so, field 3 may *not* also be specified, or the PGN will not be accepted. The 16-bit field 1 must be specified first in the list of field number and parameter value pairs in PGN 126208.

See also related PGNs:

- the received ISO Request PGN 59904
- the received Command Group Function PGN 126208
- the transmitted Acknowledge Group Function PGN 126208

NMEA 2000® RECEIVED PARAMETER GROUP

PGN 126208 (0x1ED00)

Command Group Function

Single Frame: No Access Level: 0

Field #	Name	DD	DF	Type	Resolution	Value	Remark
1	Group Function Code	DD144	DF53	uint8	1 bit	0 = Request Message 1 = Command Message 2 = Acknowledge Message 3 = Read Fields 4 = Read Fields Reply 5 = Write Fields 6 = Write Fields Reply	For the Command Group Function, this field = 1.
2	Commanded PGN	DD009	DF52	24 bits	1	Variable	24-bit Parameter Group Number (PGN) expressed in binary, LSB is transmitted first.
3	Priority Setting	DD182	DF52	4 bits	1	0x0 to 0x7 = commanded priority value 0x8 = do not change priority 0x9 = return priority to default 0xA to 0xF = reserved	
4	Reserved bits	DD001	DF52	4 bits	1	0xF	All bits set to logic "1"
5	Number of Pairs of Commanded Parameters to follow	DD006	DF53	uint8	1 bit	0 to 252	
6	Field number of first commanded parameter	DD005	DF53	uint8	1 bit	0 to 252	Field number within the Commanded PGN.
7	Value of first commanded parameter	DD000	DF00	undef.	undefined	undefined	Requested parameter size, type, and range of legal values are dependent on the Commanded PGN and the specific commanded parameter field.
8	Variable number of fields, field 6 repeated	DD005	DF53	uint8	1 bit	0 to 252	
9	Variable number of fields, field 7 repeated	DD000	DF00	undef.	undefined	undefined	

Notes

This received PGN is used by the bootloader to set the value of one or more parameters in a Parameter Group. The individual parameters of the Commanded PGN are specified in the Field Number/Value pairs (fields 6 and 7 of this PGN). Refer to the detailed description for each transmitted PGN for specific instructions regarding which fields to specify when using the Command Group Function PGN 126208. (Note that some required fields are not changeable, but are used as qualifiers when changing other fields.)

This PGN may also be used to specify the Priority field of a given transmitted PGN. (Note: certain PGNs do not permit changing the Priority field.)

Upon receiving this PGN, the device will respond by transmitting the Acknowledge Group Function PGN 126208.

All parameter value fields must be padded (with 1's) if necessary to ensure byte boundaries are adhered to.

When the Commanded PGN field 2 (of PGN 126208) refers to a proprietary PGN, fields 1 and 3 (manufacturer code and industry group) of the proprietary PGN must be specified *first* in the list of field number and parameter value pairs in PGN 126208. All other field number and parameter value pairs may appear in any order in this message.

For proprietary PGNs described in this document, fields 1, 2, and 3 (manufacturer code, reserved bits, and industry group) of the proprietary PGN may optionally be combined into a single 16-bit field when specifying fields in PGN 126208. For Airmar proprietary PGNs, the 16-bit field is constructed as follows:

- bits 15-13 = binary 100 = decimal 4 (industry group for "Marine Industry")
- bits 12-11 = binary 11 = reserved bits
- bits 10-0 = binary 000 1000 0111 = decimal 135 (manufacturer code for Airmar Technology)

The concatenation of these 16 bits produces the binary value 1001 1000 1000 0111 = hexadecimal 0x9887. Therefore, instead of specifying the two fields, 11-bit field 1 and 3-bit field 3 for a proprietary PGN, it is permissible to specify a single 16 bit field 1, containing the value 0x9887, for the proprietary PGNs described in this document.

When doing so, field 3 may *not* also be specified, or the PGN will not be accepted. The 16-bit field 1 must be specified first in the list of field number and parameter value pairs in PGN 126208.

See also related PGNs:

- the received Request Group Function PGN 126208
- the transmitted Acknowledge Group Function PGN 126208

NMEA 2000® RECEIVED PARAMETER GROUP

PGN **126720** (0x1EF00)

Addressable Multi-Frame Proprietary

Single Frame: No

Access Level: See PID variants

Field #	Name	DD	DF	Type	Resolution	Value	Remark
1	Manufacturer Code	DD172	DF52	11 bits	1	135 (decimal)	Manufacturer Code for Airmar Technology.
2	Reserved bits	DD001	DF52	2 bits	1	0x3	All bits set to logic "1"
3	Industry Group	DD168	DF52	3 bits	1	4 (decimal)	Industry Group for "Marine Industry".
4	Proprietary ID (PID)	DD005	DF53	uint8	1 bit	0 to 252	Identifier specifying how to interpret the remaining fields in this parameter group.
5+	(PID-dependent)	DD000	DF00	undef.	undefined	undefined	The use of the remaining bytes and the number of CAN frames required varies according to the PID (field 4).

Notes

In order for this received PGN to be recognized, the Manufacturer Code and Industry Group (fields 1 and 3) must match the values shown.

The use of this PGN with the following Proprietary ID's (field 4) is described on the subsequent pages:

- 1: Master Reset
- 2: Launch bootloader
- 3: Launch application
- 10: Bootloader command
- 11: Bootloader data

NMEA 2000® RECEIVED PARAMETER GROUP

PGN 126720-1 (0x1EF00)

Proprietary: Master Reset

Single Frame: No Access Level: 1

Field #	Name	DD	DF	Type	Resolution	Value	Remark
1	Manufacturer Code	DD172	DF52	11 bits	1	135 (decimal)	Manufacturer Code for Airmar Technology.
2	Reserved bits	DD001	DF52	2 bits	1	0x3	All bits set to logic "1"
3	Industry Group	DD168	DF52	3 bits	1	4 (decimal)	Industry Group for "Marine Industry".
4	Proprietary ID (PID)	DD005	DF53	uint8	1 bit	1 = Master Reset	Identifier specifying how to interpret this PGN.
5	Reserved bits	DD001	DF52	24 bits	1	0xFFFFFFFF	All bits set to logic "1"

Notes

Upon receiving this PGN, the device will perform a master CPU reset, as though the power had been turned off and back on again. This PGN performs the same function as the NMEA 0183 proprietary command, \$PAMTC,RESET.

In order for this received PGN to be recognized, the Manufacturer Code, Industry Group, and Proprietary ID (fields 1, 3, and 4) must match the values shown.

Although this is a fast-packet message, the version with this PID fits into a single CAN frame.

PGN 126720-2 (0x1EF00)

Proprietary: Launch Bootloader

Single Frame: No Access Level: 5

Field #	Name	DD	DF	Type	Resolution	Value	Remark
1	Manufacturer Code	DD172	DF52	11 bits	1	135 (decimal)	Manufacturer Code for Airmar Technology.
2	Reserved bits	DD001	DF52	2 bits	1	0x3	All bits set to logic "1"
3	Industry Group	DD168	DF52	3 bits	1	4 (decimal)	Industry Group for "Marine Industry".
4	Proprietary ID (PID)	DD005	DF53	uint8	1 bit	2 = Launch Bootloader	Identifier specifying how to interpret this PGN.
5	Reserved bits	DD001	DF52	24 bits	1	0xFFFFFFFF	All bits set to logic "1"

Notes

If the unit is running in the Startup Monitor and has already claimed a CAN address, then upon receiving this PGN, the internal flash bootloader program will be launched.

In order for this received PGN to be recognized, the Manufacturer Code, Industry Group, and Proprietary ID (fields 1, 3, and 4) must match the values shown.

Although this is a fast-packet message, the version with this PID fits into a single CAN frame.

NMEA 2000® RECEIVED PARAMETER GROUP

PGN 126720-3 (0x1EF00)

Proprietary: Launch Application

Single Frame: No Access Level: 0

Field #	Name	DD	DF	Type	Resolution	Value	Remark
1	Manufacturer Code	DD172	DF52	11 bits	1	135 (decimal)	Manufacturer Code for Airmar Technology.
2	Reserved bits	DD001	DF52	2 bits	1	0x3	All bits set to logic "1"
3	Industry Group	DD168	DF52	3 bits	1	4 (decimal)	Industry Group for "Marine Industry".
4	Proprietary ID (PID)	DD005	DF53	uint8	1 bit	3 = Launch Application	Identifier specifying how to interpret this PGN.
5	Reserved bits	DD001	DF52	24 bits	1	0xFFFFFFFF	All bits set to logic "1"

Notes

If the unit is running in the bootloader, or it is running in the Startup Monitor and has already claimed a CAN address, then upon receiving this PGN, the program stored in application flash memory will be launched.

In order for this received PGN to be recognized, the Manufacturer Code, Industry Group, and Proprietary ID (fields 1, 3, and 4) must match the values shown.

Although this is a fast-packet message, the version with this PID fits into a single CAN frame.

NMEA 2000® RECEIVED PARAMETER GROUP

PGN 126720-10 (0x1EF00)

Proprietary: Bootloader Command

Single Frame: No Access Level: 5

Field #	Name	DD	DF	Type	Resolution	Value	Remark
1	Manufacturer Code	DD172	DF52	11 bits	1	135 (decimal)	Manufacturer Code for Airmar Technology.
2	Reserved bits	DD001	DF52	2 bits	1	0x3	All bits set to logic "1"
3	Industry Group	DD168	DF52	3 bits	1	4 (decimal)	Industry Group for "Marine Industry".
4	Proprietary ID (PID)	DD005	DF53	uint8	1 bit	10 (decimal) = Bootloader Command	Identifier specifying how to interpret this PGN.
5	SID (Sequence ID)	DD056	DF53	uint8	1 bit	0 to 252	Unit-less number.
6	Command	DD005	DF53	uint8	1 bit	<u>Command codes:</u> 0 = Lock flash memory 1 = Unlock flash memory for erase/program 2 = Erase all application flash memory 3 = Program flash memory 4 = Verify flash memory 5 = Abort Program or Verify operation 6 = Change processor region 7 to 253 = reserved 254 = Out of range 255 = Do not change	<u>Serial Bootloader Equivalent:</u> 0 = (None) 1 = UNLOCK 2 = E 3 = P 4 = V 5 = <ctrl-C> 6 = R
7	Data field	DD005	DF53	uint8	1 bit	0 to 252 253 = reserved 254 = Out of range 255 = Do not change	If field 6 (Command) = 6, then this field will contain the code for the processor region being changed to. Otherwise, it should be set to 0xFF.

Notes

In order for this received PGN to be recognized, the Manufacturer Code, Industry Group, and Proprietary ID (fields 1, 3, and 4) must match the values shown.

After receiving this PGN and executing the command specified in field 6, the bootloader will transmit the proprietary PGN 126720-6, Bootloader Status. If field 6 contains the value 255, no operation will be performed, but PGN 126720-6 will be transmitted; this can be used as a shortcut method to request transmission of the Bootloader Status.

Although this is a fast-packet message, the version with this PID fits into a single CAN frame.

PRELIMINARY - DRAFT

NMEA 2000® RECEIVED PARAMETER GROUP

PGN 126720-11 (0x1EF00)

Proprietary: Bootloader Data

Single Frame: No Access Level: 5

Field #	Name	DD	DF	Type	Resolution	Value	Remark
1	Manufacturer Code	DD172	DF52	11 bits	1	135 (decimal)	Manufacturer Code for Airmar Technology.
2	Reserved bits	DD001	DF52	2 bits	1	0x3	All bits set to logic "1"
3	Industry Group	DD168	DF52	3 bits	1	4 (decimal)	Industry Group for "Marine Industry".
4	Proprietary ID (PID)	DD005	DF53	uint8	1 bit	11 (decimal) = Bootloader Data	Identifier specifying how to interpret this PGN.
5	SID (Sequence ID)	DD056	DF53	uint8	1 bit	0 to 252	Unit-less number.
6	Data	N/A	DF50	char8(n)	1 char	Variable length ASCII string, maximum 250 characters (252 bytes).	First byte in this field is the Count byte, indicating the number of bytes in the string, including the Count and Control bytes. Second byte is the Control byte, which for an ASCII string contains the value 1. The remaining bytes constitute a single S-record, starting with the ASCII character 'S' and ending with the checksum pair. No <CR> or <LF> is necessary, nor is a string terminator.

Notes

In order for this received PGN to be recognized, the Manufacturer Code, Industry Group, and Proprietary ID (fields 1, 3, and 4) must match the values shown.

After receiving this PGN, the bootloader will transmit the proprietary PGN 126720-6, Bootloader Status.

6. Serial Bootloader

The serial bootloader includes a command interpreter that communicates with the host computer using a full-duplex RS-485 interface. This is the same interface used by the application program for its NMEA 0183 IN and OUT channels. The commands recognized by the serial bootloader are separate from and uniquely different from the commands recognized by the application program, and do not follow NMEA 0183 conventions.

While the serial bootloader is executing, the CAN (NMEA 2000®) bus interface on the device is non-operational.

After power-up or reset, the RS-485 interface for the serial bootloader operates at 4800 baud, with 8 data bits, 1 stop bit, and no parity. The baud rate may be increased if desired (using the Z command).

All of the commands are a single ASCII character except the UNLOCK command, and the R and Z commands. In every case, the command characters received by the bootloader will be echoed back to the sender. After all of the characters of a recognized command have been received and echoed, the bootloader will then transmit a <CR><LF> pair.

The characters in received S-records are not echoed, but after each S-record is received a single character is transmitted indicating success or failure of the reception and programming of that S-record.

The host computer must support the XON/XOFF handshaking protocol in its transmission of S-records. The bootloader may at any time during the transmission of an S-record send an XOFF (ASCII <ctrl-S>) to the host. The host must then suspend transmission until it receives an XON (ASCII <ctrl-Q>) from the bootloader.

The following pages detail the functionality of each of the commands recognized by the serial bootloader.

The commands recognized by the serial bootloader are summarized below:

<u>Command</u>	<u>Description</u>
<ctrl-X>	Perform a soft Reset
A	Query application firmware version
B	Query bootloader firmware version
C	Query Product Code and Product Name
E	Erase Flash memory
P	Program flash
R	Change processor region
Q	Query to see if a previous flash update operation completed successfully.

<u>Command</u>	<u>Description</u>
UNLOCK	Prepare the bootloader for an Erase or Program operation
V	Verify Flash
Z	Change baud rate

IMPORTANT NOTE: The bootloader commands are *case-sensitive*. The commands are all in upper-case letters. The corresponding lower-case alternatives are *not* recognized by the bootloader.

<ctrl-X>

Purpose

Perform a soft Reset on the processor. This will cause the unit to enter the Startup Monitor.

Syntax

`<ctrl-X>` (hex 0x18)

Argument

None.

Action

Upon receiving a <ctrl-X> at the bootloader prompt, the processor will echo back the <ctrl-X> to the sender, and transmit a <CR><LF> pair. It will then perform a soft Reset and enter the Startup Monitor. This will cause the initial 2-second timeout interval to start.

If during the 2-second interval a second <ctrl-X> is received, the Startup Monitor shall then execute the application program.

If during the 2-second interval a <ctrl-B> is received (0x02), the Startup Monitor shall then execute the bootloader.

If the 2-second interval completes without receiving a <ctrl-X> or <ctrl-B>, the unit shall enter the application program.

Reply

None.

Note

In the bootloader, the <ctrl-X> is only recognized when at the bootloader prompt "01>".

A

Purpose

Report the version of Application firmware currently residing in flash memory.

Syntax

A

Argument

None.

Action

Upon receiving the 'A' command, the bootloader shall echo the received 'A', and transmit a <CR><LF> pair. It shall then transmit a 4-digit ASCII hexadecimal string representing the version of application firmware currently residing in flash memory. An implied decimal point is located 3 decimal digits from the right.

Reply

vvvv<CR><LF>

where

vvvv is the 4-digit firmware version in ASCII hex

Example:

An application with a version number of 1.002 would be stored as 1002 decimal, or 03EA hexadecimal:

Command	Reply
A	A<CR><LF> 03EA<CR><LF>

B

Purpose

Report the bootloader version.

Syntax

B

Argument

None.

Action

Upon receiving a 'B', the bootloader shall echo the received 'B', and transmit a <CR><LF> pair. It shall then transmit the version number of the Bootloader firmware (4 ASCII hex digits, representing a 16-bit value), followed by a terminating <CR><LF>. An implied decimal point is located 3 decimal digits from the right.

Reply

vvvv<CR><LF>

where

vvvv is the 4-digit bootloader version in ASCII hex

Example:

A bootloader with a version number of 1.003 would be stored as 1003 decimal, or 03EB hexadecimal:

Command	Reply
B	B<CR><LF> 03EB<CR><LF>

C

Purpose

Report Product Code string.

Syntax

C

Argument

None.

Action

Upon receiving a 'C', the bootloader shall echo the received 'C', and transmit a <CR><LF> pair. It shall then respond with the unique Airmar Product Code corresponding to this product (4 ASCII hex digits, representing a 16-bit value), followed by a terminating <CR><LF>.

Reply

cccc<CR><LF>

where

cccc is the 4-digit Product Code in ASCII hex.

Example:

Command	Reply
C	C<CR><LF> 0004 <CR><LF>

Note

The Product Code for the PB200 WeatherStation is 0004.
The Product Code for other products is TBD.

E

Purpose

Erase the flash memory.

Syntax

E

Argument

None.

Action

Upon receiving the 'E' command, the bootloader shall respond with the following prompt:

ERASE?

The host must then respond with a 'Y'. If any other character besides 'Y' is received, the bootloader shall exit to the standard prompt. After the 'Y' is received, the bootloader shall erase the flash memory containing the application program. The bootloader itself shall not be erased.

Once the flash memory has been erased, the bootloader shall respond with 'ERASED' and shall return to the standard prompt. If for some reason the bootloader is unable to erase the flash memory, the bootloader shall respond with 'Erase Error' and shall return to the standard prompt.

Example:

Command	Reply
E	E<CR><LF> ERASE? <CR><LF>
Y	Y<CR><LF> ERASED<CR><LF>

Note

The E command will not be recognized until the UNLOCK command has been successfully completed. If the UNLOCK command has not yet been performed, the E command will return with the response, "CAN'T ERASE<CR><LF>".

P

Purpose

Program the flash memory.

Syntax

P

Argument

None.

Action

Upon receiving the 'P' command, the bootloader shall echo the received 'P', and then transmit a <CR><LF> pair. It shall then expect to read a sequence of Motorola S-records from the host. The S-records shall be transmitted by the host one at a time. After each S-record is transmitted, the host shall wait for the bootloader to respond with a single character. The single character response from the bootloader will be one of the following:

- '.' (ASCII period) – the record was successfully received
- '_' (ASCII underscore) – the record was successfully received, and a sector consisting of a group of S-records was successfully programmed into flash memory
- 'X' (upper-case x) – an error occurred in the reception of the record
- 'x' (lower-case X) – an error occurred in programming the flash sector, or in the verify operation
- Any other single character response also indicates an error occurred.

The data S-records must follow the Motorola S2 format, depending on the addressing requirements of the processor being programmed.

After all data records have been transmitted, the host must transmit a single termination record (S9 or S8). The bootloader shall then respond with one of the following messages:

PROGRAM OK<CR><LF>

or

PROGRAM FAIL<CR><LF>

The bootloader shall then return to its standard prompt.

Notes

The P command will not be recognized until the UNLOCK command has been successfully completed. If the UNLOCK command has not yet been performed, the P command will return with the response, “CAN’T PROGRAM<CR><LF>”.

The P command should not be performed unless the flash memory has been previously erased with the E command.

If the bootloader response after receiving an S-record is a period ‘.’ or underscore ‘_’, then the host can proceed with transmitting the following S-record.

If the bootloader response is an uppercase 'X', then the host should retry sending the most recent S-record. If that is not successful, the programming operation should be terminated, the flash memory should be erased, and the programming operation restarted from the beginning.

If the bootloader response is a lowercase 'x', then there is no benefit to retrying the most recent S-record; in this case it is necessary to terminate the programming operation, erase the flash memory, and restart the programming operation from the beginning.

In the event the serial port input buffer on the bootloader becomes full during the transmission of S-records, the bootloader will send an XOFF (ASCII <ctrl-S>) character to the host. The host must then suspend transmission until it receives an XON (ASCII <ctrl-Q>) from the bootloader.

The programming operation may be terminated prematurely by sending a <ctrl-C> to the bootloader. This <ctrl-C> must be received between S-records, i.e. when the bootloader is waiting for the “S” character at the start of the S-record. After receiving a <ctrl-C>, the bootloader will return to its standard prompt.

Note that the S-records acceptable to the bootloader follow an encrypted, proprietary format. This format must be followed exactly, and the S-record data provided to the bootloader in the original order, to ensure that the flash memory is programmed correctly.

Q

Purpose

Query to see if a previous flash update operation completed successfully.

Syntax

Q

Argument

None.

Action

Upon receiving the 'Q' command, the bootloader shall echo the received 'Q', and then transmit a <CR><LF> pair. It shall then check the contents of a certain pair of 16-bit memory locations in the application region in flash memory. These two locations are each expected to contain a specific magic number. If these locations agree with the expected values, then the response to this command will be "Flash OK<CR><LF>". If either of these locations contain a different value than expected, then the response to this command will be "Flash Bad<CR><LF>".

Notes

A correctly formatted .ax file specifies the order in which these locations are written to flash memory during a programming operation (bootloader P command). One of the two magic numbers is the very first data word programmed to flash memory. The other magic number is the very last data word programmed to flash memory. In the event a flash programming operation is prematurely terminated (e.g. by power failure), some of the flash memory will not have been programmed properly. In particular, the second magic number will not yet have been programmed. This flash location will contain the default erased value, which will be detected as a fault by the bootloader Q command.

Note that on power up, the bootloader automatically checks the magic numbers prior to vectoring to the application program. This is independent of the Q command. If either magic number is incorrect, the unit will *not* execute the application program, but will instead enter the Startup Monitor and wait indefinitely for a command on either the NMEA 2000® interface or the serial interface.

R

Purpose

Change to a different processor region.

Syntax

R1 (Master CPU)
or
R2 (Slave CPU)

Argument

None.

Action

Upon receiving the 'R' command, the bootloader will echo both characters of the received command to the sender, followed by a <CR><LF> pair. It will then attempt to establish communication with the processor indicated by the second character of the command, and if successful, will change the bootloader prompt accordingly.

The bootloader prompt appears as follows:

Master CPU: 01>
Slave CPU: 02>

All subsequent bootloader commands received by the device will be executed by the bootloader in the CPU indicated by the prompt.

If the bootloader is unsuccessful in establishing communication with the specified processor, then the device will reply with "NO RESPONSE", and will return to the previous bootloader prompt.

Notes

If the second character is anything except 1 or 2, the command will be aborted and the unit will return to the previous bootloader prompt.

UNLOCK

Purpose

Unlock the bootloader to allow subsequent use of the E and P commands.

Syntax

UNLOCK<CR>

Argument

None.

Action

Prior to receiving the UNLOCK<CR> command, any attempt at using the E or P commands will result in an error message being produced. Upon receiving the UNLOCK<CR> command, the bootloader shall respond with the UNLOCKED<CR><LF> reply. Subsequent use of the E and P commands shall then work properly.

All of the characters of the UNLOCK<CR> command are echoed back to the sender as they are received, including the final <CR>. After having echoed the <CR>, the bootloader will then also transmit a <LF>.

Reply

UNLOCKED<CR><LF>

Example:

Command	Reply
UNLOCK<CR>	UNLOCK<CR><LF> UNLOCKED<CR><LF>

Note

The individual characters of the UNLOCK<CR> command must be received in their proper order, and terminated by a <CR> character. If any other character is received before the UNLOCK<CR> command is completed, the command is immediately aborted, and the bootloader will transmit a <CR><LF> pair. The UNLOCK<CR> command must then be retransmitted to the bootloader from the beginning.

Once the UNLOCK<CR> command has been successfully executed, the bootloader will remain unlocked until the processor is reset, either by a power cycle, or by the <ctrl-X> command.

V

Purpose

Verify that the flash memory agrees with the S-record data.

Syntax

V

Argument

None.

Action

Upon receiving the 'V' command, the bootloader shall echo the received 'V', and then transmit a <CR><LF> pair. It shall then expect to read a sequence of S-records from the host. The S-records shall be transmitted by the host one at a time. As the bootloader receives each S-record, it will compare the contents of the S-record with the contents of flash memory, and then respond with a single character. After transmitting each S-record, the host should wait for this single character response from the bootloader. The bootloader response will be one of the following:

- '.' (ASCII period) – the record was successfully received, compared against flash memory, and verified to agree
- any other single character response indicates that an error occurred either in the reception of the data record, or in the verify operation

The data S-records must follow the S2 format. They will be interpreted in exactly the same way as in the P command.

After all data records have been transmitted, the host must transmit a single termination record (S9 or S8). The bootloader shall then respond with one of the following three messages:

FILE BAD<CR><LF>

The S-record file was improperly formatted, or it produced a checksum error.

FILE OK, VERIFY FAIL<CR><LF>

The S-record file had no errors, but the flash memory did not agree with the file.

VERIFY OK<CR><LF>

The S-record file had no errors, and the flash memory agrees with the S-record file.

The bootloader shall then return to its standard prompt.

Notes

In the event the serial port input buffer on the bootloader becomes full during the transmission of S-records, the bootloader will send an XOFF (ASCII <ctrl-S>) character to the host. The host must then suspend transmission until it receives an XON (ASCII <ctrl-Q>) from the bootloader.

The verify operation may be terminated prematurely by sending a <ctrl-C> to the bootloader. This <ctrl-C> must be received between S-records, i.e. when the bootloader is waiting for the “S” character at the start of the S-record. After receiving a <ctrl-C>, the bootloader will return to its standard prompt.

Note that the S-records acceptable to the bootloader follow an encrypted, proprietary format. This format must be followed exactly, and the S-record data provided to the bootloader in the original order, to ensure that the flash memory is verified correctly.

Z

Purpose

Change the bootloader baud rate.

Syntax

Z0	(4800 baud)
Z1	(9600 baud)
Z2	(19200 baud)
Z3	(38400 baud)
Z4	(57600 baud)
or	
Z5	(115200 baud)

Argument

None.

Action

Upon receiving the 'Z' command, the Master CPU bootloader will read and echo the subsequently received character, and will transmit a <CR><LF> pair, all at the old baud rate. Then, the Master CPU will change the RS-485 interface to the new baud rate, and enter the Startup Monitor. This will cause the initial 2-second timeout interval to start.

If during the 2-second interval a <ctrl-B> is received (0x02) at the new baud rate, the Startup Monitor shall then execute the bootloader, with the bootloader prompt being provided at the new baud rate. All subsequent communication will occur at the new baud rate.

Notes

The 'Z' command is only recognized when the bootloader is at the Master CPU prompt ("01> "). It is ignored when the prompt is for the Slave CPU ("02> ").

If the second character is anything except 0, 1, 2, 3, 4, or 5, the command will be aborted, a <CR><LF> pair will be transmitted, and the unit will return to the bootloader prompt.

The baud rate is only changed for the current session. It will revert to the default rate of 4800 baud after cycling power or performing a Reset operation.

7. Flash Update Procedure: Serial

This procedure will use the PB200 WeatherStation as an example.

The recommended procedure to perform a flash update operation using a host computer communicating with the serial bootloader in the PB200 WeatherStation is as follows:

1. Set the baud rate on the host computer serial interface to 4800 baud.
2. Reset the target device either by cycling power, or, if the device is running the PB200 WeatherStation application program, by sending it the reset command:
`$PAMTC, RESET.`
3. Wait at least 0.25 seconds, but no longer than 2 seconds, to allow the Master CPU to reset and enter the Startup Monitor, then send the device a <ctrl-B> (0x02) to launch the serial bootloader. The bootloader will respond with a sign-on message and the Master CPU bootloader prompt: '01> '.
4. (Optional) Increase the baud rate using the Z command as follows:
 - a. Assuming (for example) you wish to change the baud rate to 57600 baud, transmit the command 'Z4' (at 4800 baud) to the device. The target device will re-enter the Startup Monitor, at the new baud rate.
 - b. Wait at least 0.25 seconds, but no longer than 2 seconds.
 - c. Change the baud rate on the host computer serial interface to the new baud rate (e.g. 57600 baud).
 - d. Send the device a <ctrl-B> (0x02) (at 57600 baud) to restart the serial bootloader at the new baud rate. The bootloader will respond with the bootloader prompt: '01> '.
5. Send the following commands (in any order) to the bootloader: A, B, and C. These will respond with the current Application firmware version, Bootloader firmware version, and Product Code for the Master CPU in the target device, respectively.
6. Open the .ax file to be transmitted to the target device, and parse the block header to read the firmware version, minimum bootloader version, product code, and processor code for the block. (A detailed description of the block header format can be found in section 9.)
7. Confirm that the product code in the block header agrees with the product code reported in the target device's reply to the C command. If these do not agree, then the .ax file was intended for a different product, and should not be loaded into this product.
8. Confirm that the processor code in the block header agrees with the processor code provided by the bootloader prompt. If these do not agree, then use the R command to instruct the bootloader to change to the processor region specified in the block header.

9. Confirm that the bootloader version reported in the target device's reply to the B command is greater than or equal to the minimum bootloader version reported in the block header. If it is not, then this file should not be loaded into the target device.
10. (Optional) Report to the user the current application firmware version in the device (reported in the reply to the A command), along with the firmware version for the new file (from the block header), and give the user an opportunity to confirm whether to proceed with the flash update procedure.
11. Send the UNLOCK<CR> command to the target device.
12. Send the 'E' command to the target device, followed by a 'Y' in response to the ERASE? query.
13. Send the 'P' command to the target device to initiate the programming operation.
14. Send the entire block to the target device, including the block header, all data records in their original order, and the termination record. While the block is being transmitted, monitor the serial port for any received XOFF and XON characters (<ctrl-S>, 0x13 and <ctrl-Q>, 0x11, respectively) to control the pacing of characters and prevent input buffer overflow on the target device. After each S-record is transmitted, the bootloader will respond with a single character indicating the success or failure in reading or programming that S-record: a period ('.', 0x2E) or underscore ('_', 0x5F) indicates success; any other character indicates failure.
15. If it is necessary to abort the procedure, transmit a <ctrl-C> (0x03) to the bootloader. (Note: a <ctrl-C> will only be accepted when the bootloader is expecting the start of the next S-record.) The device will respond with the bootloader prompt '01> '.
16. After the termination record is received by the bootloader, it will respond with one of the following messages: 'PROGRAM OK', or 'PROGRAM FAIL', followed by the bootloader prompt. If PROGRAM FAIL is transmitted, then the Erase/Program operation will need to be repeated, perhaps at a slower baud rate.
17. After the block has been successfully programmed, then read the block header for the next block from the .ax file, change to the corresponding processor region using the 'R' command, and repeat the above procedure for the next processor region. This should be done for each block in the .ax file.
18. When finished, change back to the Master CPU bootloader prompt with the R1 command (if necessary), and then transmit a <ctrl-X> to the bootloader to reset the Master CPU, which will launch the Startup Monitor. Optionally, a second <ctrl-X> may then be transmitted to the Startup Monitor (at 4800 baud) to abort the 2-second timeout and immediately launch the application program.

8. Flash Update Procedure: NMEA 2000®

This procedure will use the PB200 WeatherStation as an example.

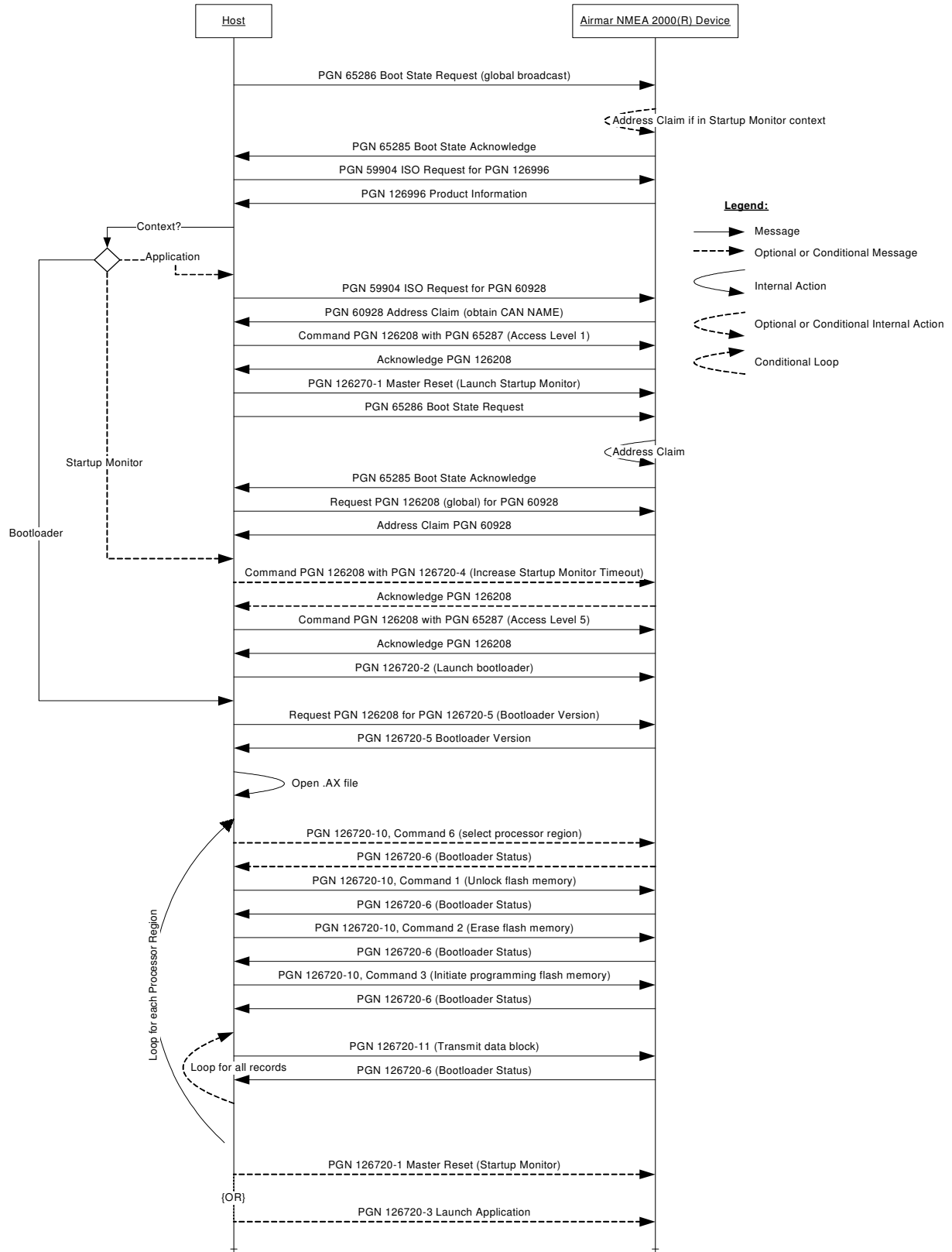
The recommended procedure to perform a flash update operation using a host computer communicating with the NMEA 2000® bootloader in the PB200 WeatherStation is as follows. Also refer to the Event State Diagram presented in Figure 2 on page 62.

1. The host starts by transmitting the global Airmar proprietary PGN 65286, Boot State Request, onto the NMEA 2000® bus. All Airmar devices connected to the bus and running in their Application, Bootloader, or Startup Monitor contexts will respond with proprietary PGN 65285, Boot State Acknowledgment. (Devices running the Startup Monitor will first perform an address claim procedure prior to transmitting PGN 65285.)
2. Then, send an ISO request to each of the responding devices for the transmission of PGN 126996, Product Information. The information provided by that PGN from each device (e.g. Model ID, Serial Number, etc.) may be displayed to the user by the host, to allow the user to select which device he wishes to perform a flash upgrade on.
3. The following steps will depend on which context the device is in: Application, Bootloader, or Startup Monitor. This context was reported in proprietary PGN 65285. If the device is already in its Bootloader context, then skip to step 13.
4. If PGN 65285 reported that the device is in its Startup Monitor context, then skip to step 10.
5. At this point, we know the device is running in its Application context. Send an ISO request to the selected device for the ISO Address Claim PGN 60928, to retrieve the device's unique CAN NAME.
6. Use the Command Group Function PGN 126208 with Airmar proprietary PGN 65287 to set the Access Level to 1 or higher.
7. Transmit the Airmar proprietary PGN 126720-1 to the device to cause it to perform a Master Reset, which will launch the Startup Monitor within the target device and start its 2-second timeout.
8. Again transmit the global Airmar proprietary PGN 65286, Boot State Request. Once again, all Airmar devices connected to the bus will respond with proprietary PGN 65285, Boot State Acknowledgment. The target device of interest will be running in the Startup Monitor, and will perform an address claim procedure prior to transmitting PGN 65285.
9. Use the Request Group Function PGN 126208, with global destination, to request transmission of the ISO Address Claim PGN 60928 by the device having the unique CAN NAME determined in step 5. This will allow the host to identify the address currently used by the target device.

10. It may be necessary to use the Command Group Function PGN 126208 with proprietary PGN 126720-4 to increase the Startup Monitor timeout in the target device.
11. Use proprietary PGN 65287 to set the Access Level to 5.
12. Transmit proprietary PGN 126720-2 to launch the bootloader in the target device.
13. While the target device is operating in its bootloader, it will be necessary for the host computer to continually monitor the NMEA 2000® bus for any address claim message activity (PGN 60928). In particular, if any device on the bus attempts to claim the address being used by the target device, the host computer must ensure that it continues to maintain the correct one-to-one correspondence between the CAN NAME and the address of the target device, before continuing further communication with the target.
14. Use the Request Group Function PGN 126208 to request proprietary PGN 126720-5, Bootloader Version, from the target device. The response from the target bootloader will include the current Application firmware version, Bootloader firmware version, Product Code, and Processor Code for the Master CPU in the target device.
15. Open the .ax file to be transmitted to the target device, and parse the block header to read the firmware version, minimum bootloader version, product code, and processor code for the block. (A detailed description of the block header format can be found in section 9.)
16. Confirm that the product code in the block header agrees with the product code reported by the target device. If these do not agree, then the .ax file was intended for a different product, and should not be loaded into this product.
17. Confirm that the processor code in the block header agrees with the processor code reported by the target device. If these do not agree, then use PGN 126720-10, command 6, to instruct the bootloader to change to the processor region specified in the block header.
18. Confirm that the bootloader version reported by the target device is greater than or equal to the minimum bootloader version reported in the block header. If it is not, then this file should not be loaded into the target device.
19. (Optional) Report to the user the current application firmware version in the device, along with the firmware version for the new file (from the block header), and give the user an opportunity to confirm whether to proceed with the flash update procedure.
20. Use PGN 126720-10, command 1, to unlock the flash memory. Verify the success of this operation by checking the reply provided by the target in the proprietary Bootloader Status PGN 126720-6.
21. Use PGN 126720-10, command 2, to erase the flash memory. Verify the success of this operation by checking the reply provided by the target in the proprietary Bootloader Status PGN 126720-6.
22. Use PGN 126720-10, command 3, to initiate the programming operation. Verify the success of this operation by checking the reply provided by the target in the proprietary Bootloader Status PGN 126720-6.

23. Using a series of transmitted PGNs 126720-11, send the entire block to the target device, including the block header, all data records in their original order, and the termination record. Each S-record must occupy a single parameter group 126720-11. The SID field must be incremented by 1 for each successively transmitted 126720-11. After the bootloader has received and processed each PGN, it will respond by transmitting the proprietary Bootloader Status PGN 126720-6, with the SID set to match the SID of the record being processed, and status field indicating 0 (success) or an appropriate error code. The host computer should check the status field to determine whether it is OK to proceed with the next S-record.
24. If it is necessary to abort the procedure, use PGN 126720-10, command 5.
25. After the termination record is received by the bootloader, the Bootloader Status PGN 126720-6 transmitted by the device will indicate in its Status field the code for Program or Verify Operation Successful, and the Programming Mode field will change from the Unlocked, Program Mode state (2) to the Unlocked state (1).
26. After the block has been successfully programmed, then read the block header for the next block from the .ax file, and repeat the above procedure from step 17 for the next processor region. This should be done for each block in the .ax file.
27. When finished, either return to the Startup Monitor by using PGN 126720-1, or launch the application by using PGN 126720-3.

Figure 2: NMEA 2000® Flash Update Procedure Event Trace Diagram



9. .AX File Format

During a flash update procedure, a host computer transfers a data file containing new firmware to the bootloader in the target device. Data files recognized by the Airmar bootloader are based on the Motorola S-record file format (described in section 10.). These files are referred to as Airmar Encoded Hex files (or '.ax files'), and have filenames ending in the extension '.ax'.

Each .ax file is composed of one or more *blocks* in sequential order. Each block contains all information necessary to upgrade the firmware in one CPU in the target device. Therefore, an .ax file containing more than one block can be used to upgrade the firmware in more than one CPU in the target device. Each block consists of the following three components, in order:

1. A single *block header* (S0 record)
2. One or more *data records* (S1, S2, or S3 records)
3. A single *termination record* (S7, S8, or S9 record)

The contents of the block header are as follows.

<u>Field</u>	<u>Characters</u>	<u>Data type</u>	<u>Description</u>
1	2	N/A	'S0'
2	2	uint8	Count of character pairs remaining in S-record
3	4	uint16	S-record address field (always '0000')
4	2	uint8	Header type (block header = 'FF')
5	2	uint8	Block format ('01')
6	2	uint8	Number of blocks in file
7	2	uint8	Sequential block number of this block
8	4	uint16	Product code ('0004' for PB200 WeatherStation)
9	2	uint8	Processor code ('01'=Master, '02'=Slave)
10	2	uint8	Reserved
11	4	uint16	Minimum bootloader version required to read this block
12	4	uint16	Firmware version
13	8	uint32	Checksum
14	4	uint16	Year timestamp
15	2	uint8	Month timestamp (1-12)
16	2	uint8	Day timestamp (1-31)
17	2	uint8	Hour timestamp (0-23)
18	2	uint8	Minute timestamp (0-59)
19	2	uint8	Second timestamp (0-59)
20	Variable	hex ASCII	Copyright message, followed by string terminator ('00')
21	Variable	hex ASCII	Product name, followed by string terminator ('00')

10. Appendix: Motorola S-Record Format

The following fairly complete description of Motorola S-records was found a number of years ago on the man pages of a UNIX system.

NOTE

- There are some inconsistencies within the following description with regard to the permissible length of an S-record.
- The following should be regarded as guidelines regarding S-records in general, and does not address specific limitations or extensions present in the Airmar flash bootloader .ax file format.

NAME

srec - S-record file and record format

DESCRIPTION

An S-record file consists of a sequence of specially formatted ASCII character strings. An S-record will be less than or equal to 78 bytes in length.

The order of S-records within a file is of no significance and no particular order may be assumed.

The general format of an S-record follows:

```
+-----//-----//-----+
| type | count | address |          data          | checksum |
+-----//-----//-----+
```

type -- A char[2] field. These characters describe the type of record (S0, S1, S2, S3, S5, S7, S8, or S9).

count -- A char[2] field. These characters when paired and interpreted as a hexadecimal value, display the count of remaining character pairs in the record.

address -- A char[4,6, or 8] field. These characters grouped and interpreted as a hexadecimal value, display the address at which the data field is to be loaded into memory. The length of the field depends on the number of bytes necessary to hold the address. A 2-byte address uses 4 characters, a 3-byte address uses 6 characters, and a 4-byte address uses 8 characters.

data -- A char [0-64] field. These characters when paired and interpreted as hexadecimal values represent the memory loadable data or descriptive information.

checksum -- A char[2] field. These characters when paired and interpreted as a hexadecimal value display the least significant byte of the ones complement of the sum of the byte values represented by the pairs of characters making up the count, the address, and the data fields.

Each record is terminated with a line feed. If any additional or different record terminator(s) or delay characters are needed during transmission to the target system it is the responsibility of the transmitting program to provide them.

S0 Record. The type of record is 'S0' (0x5330). The address field is unused and will be filled with zeros (0x0000). The header information within the data field is divided into the following subfields.

mname is char[20] and is the module name.
ver is char[2] and is the version number.
rev is char[2] and is the revision number.
description is char[0-36] and is a text comment.

Each of the subfields is composed of ASCII bytes whose associated characters, when paired, represent one byte hexadecimal values in the case of the version and revision numbers, or represent the hexadecimal values of the ASCII characters comprising the module name and description.

S1 Record. The type of record field is 'S1' (0x5331). The address field is interpreted as a 2-byte address. The data field is composed of memory loadable data.

S2 Record. The type of record field is 'S2' (0x5332). The address field is interpreted as a 3-byte address. The data field is composed of memory loadable data.

S3 Record. The type of record field is 'S3' (0x5333). The address field is interpreted as a 4-byte address. The data field is composed of memory loadable data.

S5 Record. The type of record field is 'S5' (0x5335). The address field is interpreted as a 2-byte value and contains the count of S1, S2, and S3 records previously transmitted. There is no data field.

S7 Record. The type of record field is 'S7' (0x5337). The address field contains the starting execution address and is interpreted as 4-byte address. There is no data field.

S8 Record. The type of record field is 'S8' (0x5338). The address field contains the starting execution address and is interpreted as 3-byte address. There is no data field.

S9 Record. The type of record field is 'S9' (0x5339). The address field contains the starting execution address and is interpreted as 2-byte address. There is no data field.

EXAMPLE

Shown below is a typical S-record format file.

```
S00600004844521B
S1130000285F245F2212226A000424290008237C2A
S11300100002000800082629001853812341001813
S113002041E900084E42234300182342000824A952
```

S107003000144ED492
S5030004F8
S9030000FC

The file consists of one S0 record, four S1 records, one S5 record and an S9 record.

The S0 record is comprised as follows:

- S0 S-record type S0, indicating it is a header record.
- 06 Hexadecimal 06 (decimal 6), indicating that six character pairs (or ASCII bytes) follow.
- 00 00 Four character 2-byte address field, zeroes in this example.
- 48 44 52 ASCII H, D, and R - "HDR".
- 1B The checksum.

The first S1 record is comprised as follows:

- S1 S-record type S1, indicating it is a data record to be loaded at a 2-byte address.
- 13 Hexadecimal 13 (decimal 19), indicating that nineteen character pairs, representing a 2 byte address, 16 bytes of binary data, and a 1 byte checksum, follow.
- 00 00 Four character 2-byte address field; hexadecimal address 0x0000, where the data which follows is to be loaded.
- 28 5F 24 5F 22 12 22 6A 00 04 24 29 00 08 23 7C Sixteen character pairs representing the actual binary data.
- 2A The checksum.

The second and third S1 records each contain 0x13 (19) character pairs and are ended with checksums of 13 and 52, respectively. The fourth S1 record contains 07 character pairs and has a checksum of 92.

The S5 record is comprised as follows:

- S5 S-record type S5, indicating it is a count record indicating the number of S1 records
- 03 Hexadecimal 03 (decimal 3), indicating that three character pairs follow.
- 00 04 Hexadecimal 0004 (decimal 4), indicating that there are four data records previous to this record.
- F8 The checksum.

The S9 record is comprised as follows:

- S9 S-record type S9, indicating it is a termination record.
- 03 Hexadecimal 03 (decimal 3), indicating that three character pairs follow.
- 00 00 The address field, hexadecimal 0 (decimal 0) indicating the starting execution address.
- FC The checksum.

NOTES

- There isn't any evidence that Motorola ever has made use of the header information within the data field of the S0 record, as described above. This must have been used by some third party vendors.
- This is the only place that a 78-byte limit on total record length or 64-byte limit on data length is documented. These values shouldn't be trusted for the general case.
- The count field can have values in the range of 0x3 (2 bytes of address + 1 byte checksum = 3, a not very useful record) to 0xff; this is the count of remaining character pairs, including checksum.
- If you write code to convert S-Records, you should always assume that a record can be as long as 514 (decimal) characters in length ($255 * 2 = 510$, plus 4 characters for the type and count fields), plus any terminating character(s). That is, in establishing an input buffer in C, you would declare it to be an array of 515 chars, thus leaving room for the terminating null character.

11. Revision History

Revision 0.001

Date: 9/11/06
Description: Draft for Airmar internal review and comment.

Revision 0.002

Date: 9/15/06
Description: Draft for customer review and comment.

Revision 0.003

Date: 10/12/06
Description: Draft for customer review and comment.

Revision 0.004

Date: 1/18/07
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Revision 0.005

Date: 3/7/07
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